16

The Future of an Allusion

PROSPECTS FOR SPACE AND TIME

Physicists spend a large part of their lives in a state of confusion. It's an occupational hazard. To excel in physics is to embrace doubt while walking the winding road to clarity. The tantalizing discomfort of perplexity is what inspires otherwise ordinary men and women to extraordinary feats of ingenuity and creativity; nothing quite focuses the mind like dissonant details awaiting harmonious resolution. But en route to explanation—during their search for new frameworks to address outstanding questions—theorists must tread with considered step through the jungle of bewilderment, guided mostly by hunches, inklings, clues, and calculations. And as the majority of researchers have a tendency to cover their tracks, discoveries often bear little evidence of the arduous terrain that's been covered. But don't lose sight of the fact that nothing comes easily. Nature does not give up her secrets lightly.

In this book we've looked at numerous chapters in the story of our species' attempt to understand space and time. And although we have encountered some deep and astonishing insights, we've yet to reach that ultimate eureka moment when all confusion abates and total clarity prevails. We are, most definitely, still wandering in the jungle. So, where from here? What is the next chapter in spacetime's story? Of course, no one knows for sure. But in recent years a number of clues have come to light, and although they've yet to be integrated into a coherent picture, many physicists believe they are hinting at the next big upheaval in our understanding of the cosmos. In due course, space and time as currently conceived may be recognized as mere allusions to more subtle, more profound, and more fundamental principles underlying physical reality. In the final chapter of this account, let's consider some of these clues and catch a glimpse of where we may be headed in our continuing quest to grasp the fabric of the cosmos.